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VEST-RL: Learning Bonus and Equity Fields for Efficient Multi-Agent Systems



NeurIPS 2026 Conference Submission29968 Authors

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TL;DR: VEST-RL learns graph-structured incentive fields that help multi-agent teams coordinate more efficiently under budget and fairness constraints.

Abstract:

External incentive structures often determine whether a team remains trapped in low-efficiency behavior or transitions into sustained high-productivity coordination. Yet most multi-agent learning frameworks treat the reward function as fixed and focus exclusively on learning agent policies. We introduce mechanism-conditioned multi-agent production, a setting in which an external planner learns dynamic incentive structures for a team of interacting agents. Our key object is the incentive field, a graph-structured and temporally extended mechanism that unifies individual bonuses, pairwise collaboration rewards, milestone incentives, and vesting-style delayed ownership rewards. We develop VEST-RL, a bilevel reinforcement learning framework that learns budget-aware and fairness-aware incentive fields from interaction data. Theoretically, we show that bounded separable scalar bonuses can be insufficient when local transfers cannot aggregate relational credit; that structured near-optimal incentive fields admit sparse low-rank approximation; that moderate vesting incentives can alter equilibrium selection in an aggregate coordination model; and that projected two-timescale mechanism-policy learning approaches stationary limit sets of the constrained Stackelberg objective. Empirically, on a SNAP email-Eu-core graph-grounded benchmark with a strict no-oracle protocol, VEST-RL attains higher cumulative output than scalar and reward-shaping baselines, with a matched-seed paired bootstrap confidence interval for the output gap that excludes zero; it also reports higher completion and descriptively low realized inequality under explicit budget and fairness accounting. A larger-team stress test shows the same directional output pattern without being used for scaling-law claims. A post-hoc support-ranking diagnostic reports higher mean AUPRC than one non-relational baseline; we treat threshold-level support claims and rankings among relational alternatives as outside the scope of the headline evidence. We report only numerical claims supported by graph-grounded audits and provide reproducibility protocols for broader embodied and LLM-agent extensions. Our results suggest that in multi-agent systems, learning the right institution can be as important as learning the right policy.

Supplementary Material: 📄 zip (/attachment?id=Glvffn4dqR&name=supplementary_material)
Primary Area: Decision making, reinforcement learning, and control (e.g., hierarchical RL, multi-agent systems, agentic AI)
Secondary Area: AI and network science (e.g., graph neural networks, communication), Optimization (e.g., convex and non-convex, stochastic, robust), Theory (e.g., learning theory, theory of deep learning, algorithmic game theory)
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VEST-RL studies how to learn the incentive mechanism for a multi-agent system, not just the agents’ policies. It introduces graph-structured bonuses, collaboration rewards, milestone incentives, and vesting-style rewards, and shows these can improve collective output under budget and fairness constraints.

Strengths And Weaknesses:

VEST-RL tackles an interesting and timely problem: learning the incentive structure itself, rather than only learning agent policies. The paper is strongest in its framing and in the way it combines graph-structured incentives, budget and fairness constraints, and a bilevel RL setup into one coherent method. The theoretical results are scoped but relevant, and the empirical audit is reasonably careful, with matched seeds, confidence intervals, and a clear no-oracle protocol.

The main weaknesses are that the evaluation is still fairly synthetic and benchmark-specific, and several claims are intentionally narrow. The paper is also conceptually dense, so the connection between the theory, the mechanism design choices, and the empirical diagnostics is not always easy to follow at first reading.

Quality: 3: good

Clarity: 3: good

Significance: 3: good

Originality: 4: excellent

Questions:

How much of the improvement remains if the fairness and budget penalties are tightened further? This would help judge whether the method still works under stricter constraints.

Limitations:

Yes, the authors do discuss the main limitations clearly, especially the simulator setting, the dependence on counterfactual estimation, and the fact that the structural diagnostic is not graph reconstruction. I would only suggest making the deployment scope even more explicit, especially around what would be needed to move from graph-grounded simulation to real human or LLM-agent teams.

Rating: 4: Borderline accept: Technically solid paper where reasons to accept outweigh reasons to reject, e.g., limited evaluation. Please use sparingly.

Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.

Ethical Concerns: NO or VERY MINOR ethics concerns only

Paper Formatting Concerns:

No major issues noted from the paper itself.

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes

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